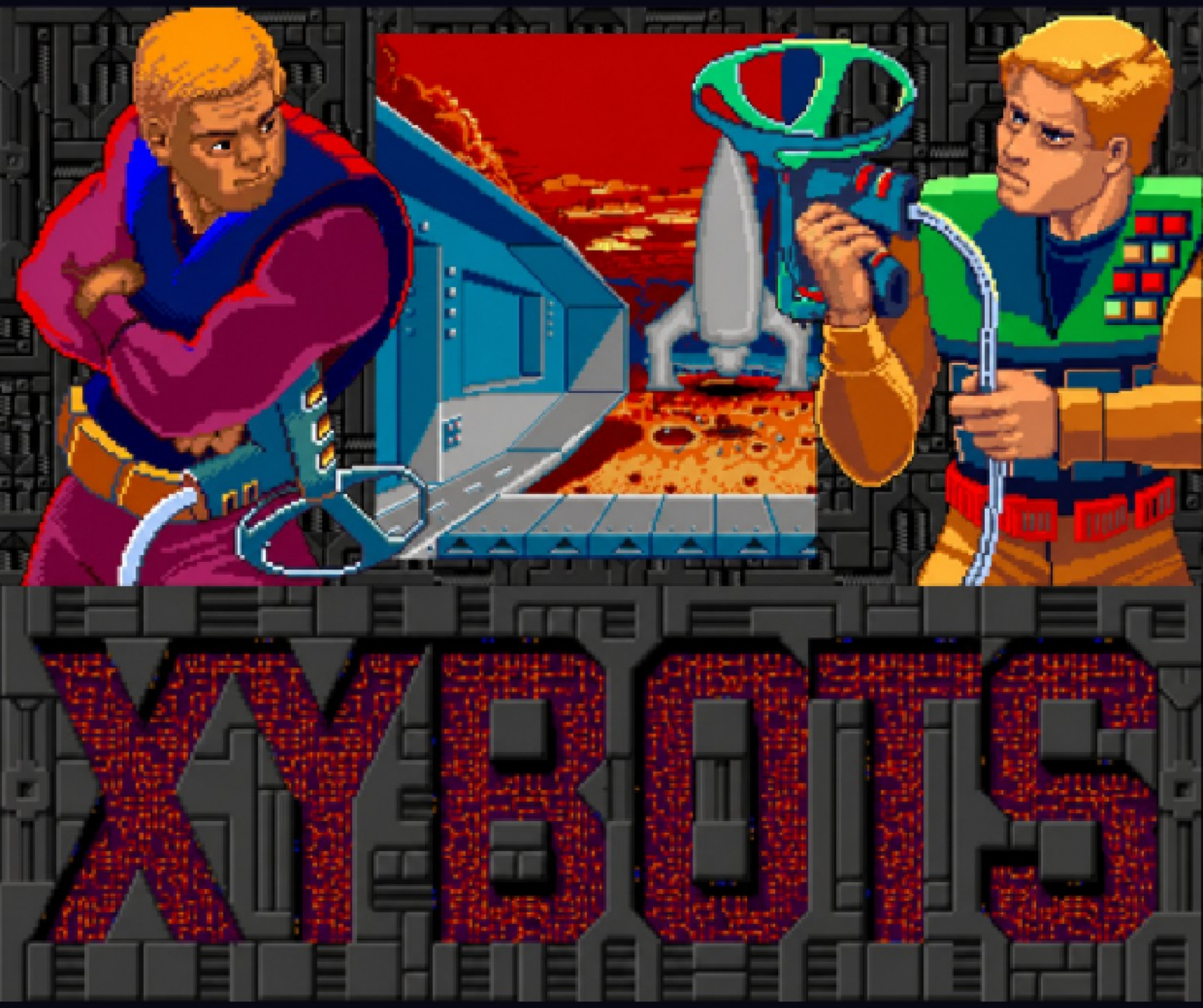


XYBOTS

MSX

V9990

USER MANUAL

The MSX game on the V9990

XYBOTS

for the Yamaha V9990

MSX V9990

An original game by **Domark (Tengen)**, 1989.
V9990 port by **DiHalt Studio**, 2026.



This is an unofficial port made for study, preservation, and homage.

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1. What you need

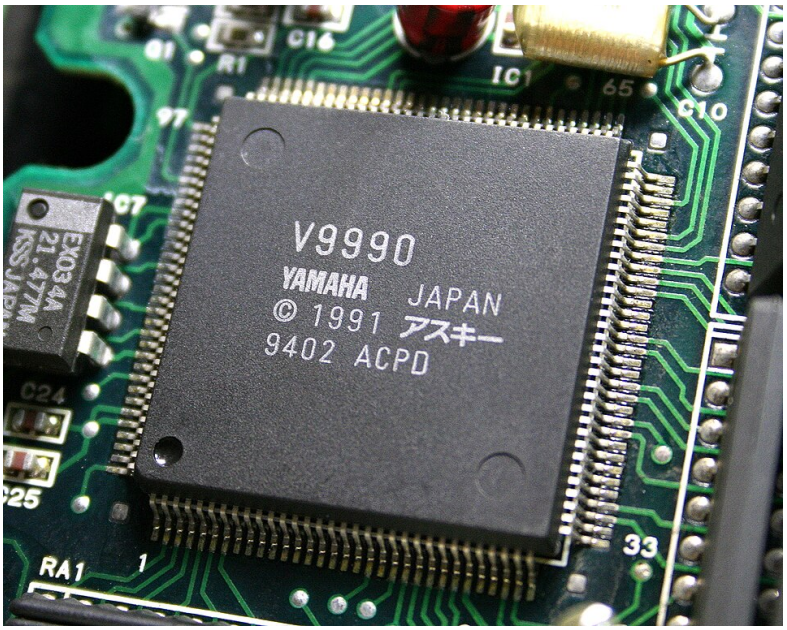


Figure 1: The Yamaha V9990 video processor, the chip this version of the game draws with.

Item	Requirement
Computer	**Any MSX1** with two cartridge slots, one for the game and one for the V9990, and a third one if you add a Moonsound. If your MSX does not have enough free cartridge slots, connect them through a **slot expander**.
Memory	**64 KB of RAM** in the main memory (all four pages).
Video	A **V9990 cartridge** (Yamaha V9990 video). It is mandatory: the whole picture is drawn by the V9990. The computer's own video chip is kept blank on purpose.
Sound	The MSX PSG is used. A **Moonsound** (OPL4) is optional: it adds arcade music and voices. With more than 128 KB of Moonsound sample RAM, some effects also use arcade sounds.
Game	The **Xybots V9990 cartridge**: a 2 MiB ASCII16 MegaROM (the file `xybots.rom`, written to a flash cartridge or equivalent).
Monitor	A monitor or TV connected to the **V9990's video output**, not to the MSX's own video output.
Controls	The keyboard, for one or two players. Joysticks are optional: port 1 for player 1, port 2 for player 2.

No tape, no disk, no password. The whole game is in the cartridge.

2. Setting up

1. **Switch the MSX off.** Never insert or remove a cartridge with the power on.
2. Insert the **V9990 cartridge** in a free slot and, if you have one, the **Moonsound** in another. If your MSX does not have enough free cartridge slots, connect them through a **slot expander**.
3. Insert the **Xybots cartridge** in another free slot. The order does not matter.
4. Connect your monitor or TV to the **video output of the V9990**.

5. Plug joysticks into port 1 (player 1) and port 2 (player 2) if you want to use them; the keyboard works for both players anyway.
6. Switch the MSX on. The DiHalt Studio logo appears on the V9990's monitor. If the MSX's own screen says `This cartridge needs a V9990.` the V9990 is missing or not seated, and the game stops there.

If the screen stays black check, in this order: the monitor is on the V9990's output; the V9990 and the game are fully inserted; the computer has 64 KB of RAM; the game ROM is the 2 MiB ASCII16 MegaROM build. On an emulator such as openMSX, enable the V9990 extension and load the ROM with the right mapper.

3. From power-on to the game

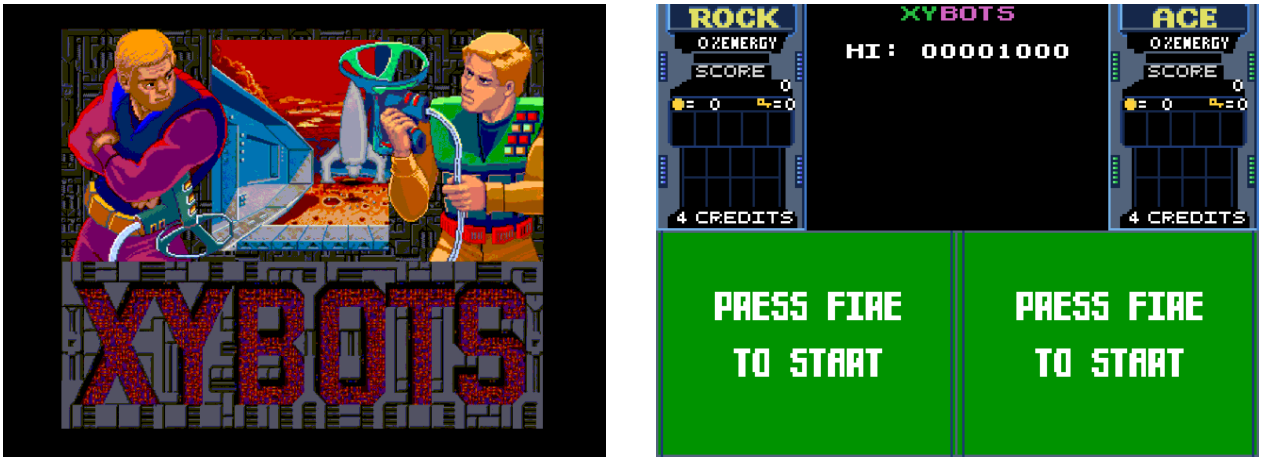


Figure 2: Left: the cover picture. Right: the title screen.

1. **DiHalt Studio logo**, 4 seconds without a Moonsound, or 10 seconds with its music and voices on a Moonsound. It cannot be skipped.
2. **The cover picture**: Rock and Ace with the `XYBOTS` title. It stays until you press **SPACE** or a joystick button. It is shown only once after switching on.
3. **The title screen**: the two views say `PRESS FIRE TO START`; above them the name of the game, the authors and `HI:` with the high score. With a Moonsound the arcade title music plays.
4. **The game**: player 1 starts with **SPACE** or the button of joystick 1, player 2 with **CTRL** or the button of joystick 2. Each player has a view of their own: Rock on the left, Ace on the right. Player 2 can join later as well (section 5).

When no player is left in the game and the credits are used up, the game goes back to the title screen (section 8).

4. Controls

Action	Player 1	Player 2	Joystick
Walk forward / back	↑ ↓	W S	up / down
Step left / right	← →	A D	left / right
Turn left / right	N M	SHIFT + A / D	button B + left / right
Fire	SPACE	CTRL	button A
Special attack	SPACE + ↓	CTRL + S	button A + down
Pause / go on	P (any player)		

- Joystick 1 (port 1) plays for player 1 and joystick 2 (port 2) for player 2, together with the keys.
- The shot leaves when you **release** the fire button. You can keep walking while you hold it.
- A turn is a quarter turn. Holding the turn key keeps turning, one quarter after another.
- Player 2 turns by holding SHIFT and pressing A or D (they do not step then). CTRL + A / D fires while stepping sideways, like SPACE + ← / → for player 1.
- The **special attack** costs energy: the view flashes red, every robot in sight is stunned and hurt, and the robots' shots in sight disappear.
- In the shop, up and down choose and fire buys (section 7). On the screens that say `PRESS FIRE`, fire goes on.

5. How to play



Figure 3: Left: Rock and Ace in the first level, each in their own view. Right: robots in level 5.

Rock (player 1) and Ace (player 2) go through the mazes of the Xybots' city, seen from behind the hero. Each level is a maze full of robots, coins, keys and energy. **Find the exit of the level and walk into it.** The level number is shown before it starts (`PREPARE TO ENTER LEVEL`).

- **Robots.** Shoot them; tougher ones need several hits. Some walk straight into you: the crash takes 25 energy. Robots can pick up objects and drop them when they are destroyed. Some walls can be shot away too.

- **Energy.** You start with 100. It goes down slowly with time, and with every shot or crash that hits you. At 0 the hero dies and a credit is used (section 8).
- **Keys** open the doors marked on the map. Walking into a locked door uses one key and the door disappears.
- **Coins** are the money of the shop between levels (section 7).
- **Two players.** Player 2 can join while the game goes on by pressing fire; the screen of Ace says `PRESS FIRE TO START`. It is not possible during the shop, bonus and waiting screens or a Master Xybot fight. From level 6 on, `SHOTS HURT OTHER PLAYERS`: take care not to hit your partner. When a player dies, the partner gets their keys.
- **After the exit** come a bonus screen (if any), the shop and the `PREPARE` screen. In a two-player game the first one out shows `WAITING...` until the partner arrives, and gets `BONUS FOR EXITING FIRST`: one coin.
- **Points**: every object picked up gives 100 points, every robot destroyed gives points as well.

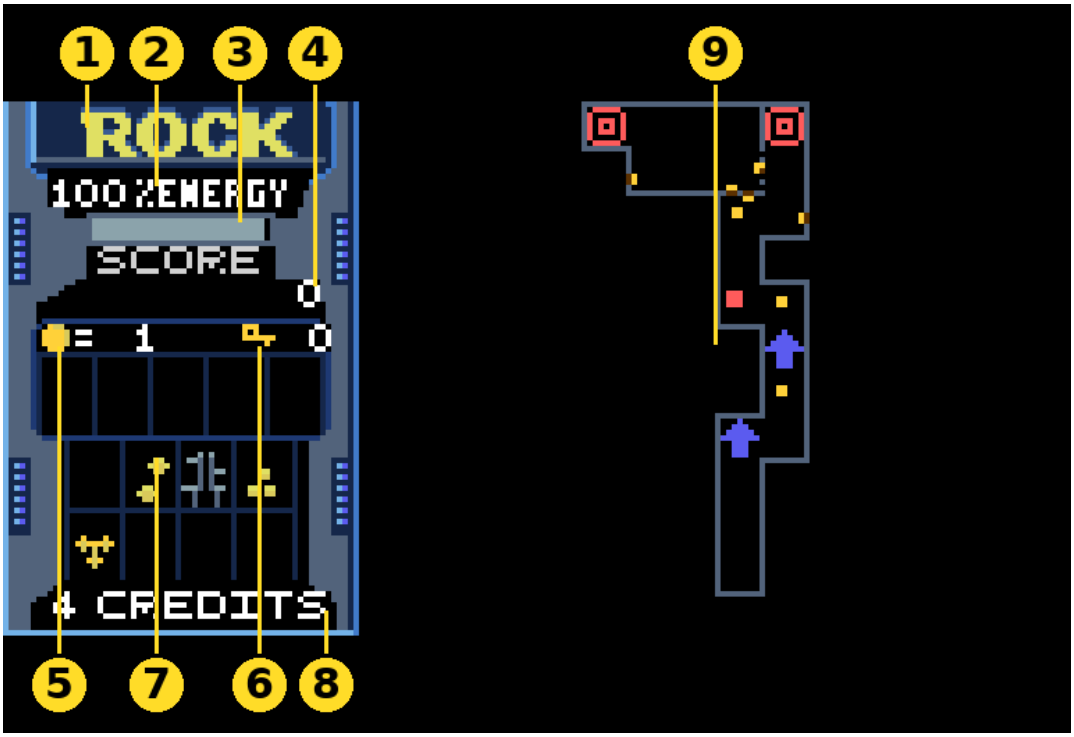


Figure 4: Left: the Master Xybot. Right: the bonus for defeating him.

The Master Xybot. Instead of levels 16, 26 and 36 you fight the Master Xybot in his room (`PREPARE TO FIGHT MASTER XYBOT`; the `PREPARE` screen warns you a few levels before). His six parts open and close and fire when they are open; they can only be hurt while open. Destroy all six: `! BONUS !` gives 10,000 points, 20 coins and full energy, and the game goes on with the next level. If every player dies in the fight, the game goes back four levels. After the last level the game starts again from level 1, with tougher robots.

6. The panel

Each player has a panel: Rock on the left, Ace on the right. The map is in the middle.



No.	Element	What it tells you
1	Name	`ROCK` is player 1, `ACE` player 2.
2	`%ENERGY`	The energy left, from 0 to 100 (125 with `125% LIMIT`).
3	Energy bar	The same energy as a bar.
4	`SCORE`	The points of the player.
5	Coins	The coins the player carries, to spend in the shop.
6	Keys	The keys the player carries.
7	Equipment	An icon for every upgrade or gadget of the shop the player owns.
8	`CREDITS`	The credits left; both players share them.
9	Map	The part of the maze seen so far: walls, locked doors (dotted), the exits (a square inside a square), the heroes (arrows that show where they look) and marks for robots and objects. It stays empty in the Master Xybot's room.

7. Objects and the shop

On the floor: coins (one coin), keys (one key), energy (50 or 100 energy, never above the limit) and the gun power-up: for a while the shots are stronger and the hero fires two balls at a time, or three when the power-up is taken again.



Figure 5: The shop: `YOU HAVE $21` coins; up and down choose, fire buys.

The shop opens after every exit when you have coins. It offers up to six things, with the number of coins each costs in front. Up and down move the bar, fire buys; `** DONE **` leaves. The shop closes by itself when your coins run out. Upgrades can be bought more than once, up to their maximum.

Item	Coins	What it does
`MOVE FASTER`	7	Walk faster.
`BETTER SHIELD`	7	Shots hurt less.
`FAST SHOTS`	7	Your shots fly faster.
`SHOT POWER`	7	Your shots hurt more.
`ZAP POWER`	7	A stronger special attack (it costs more energy too).
`SLOW ENERGY`	2	The energy goes down more slowly with time.
`SECOND SHOT`	1	Two shots in the air at a time.
`ENEMY MAPPER`	1	The robots appear on the map.
`WALL MAPPER`	1	The whole maze appears on the map.
`GUARD MAPPER`	1	The guard robots appear on the map.
`WARNING ARROWS`	1	`DANGER` signs at the side of the view when a robot can reach you straight.
`50% ENERGY POD`	5	50 energy.
`SHOTS REPELLED`	5	The next five shots that hit you bounce off.
`125% LIMIT`	5	The energy can go up to 125.
`KEY`	1	One key.
`GIVE FRIEND`	1	Gives one coin to your partner.

From level 7 on, the gadgets (second shot, mappers, warning arrows, 125% limit) wear off: every exit takes one step of each away.

8. Credits, continue and high score

- The game starts with **4 credits**, shared by both players (`4 CREDITS` in the panel).
- When a hero dies, one credit is used and the view says `PRESS FIRE TO CONTINUE`. Fire puts the hero back at the start of the level with full energy and everything bought kept.
- The death that uses the **last credit** ends that player's game: if the score beats the high score, `NEW HISCORE` flashes; then `GAME OVER`. With 4 credits a player can continue three times.

- The high score starts at 1000 and is shown on the title screen as `HI:`. It is kept until the MSX is switched off.
- Joining as player 2 costs no credit.

9. Music and voices

Without a Moonsound you hear the music and sound effects of the original MSX game. With a **Moonsound**, the music and voices are:

- a song with voices on the logo screen;
- the arcade title music with its voices on the title screen;
- one of four arcade voices, at random, when a game starts;
- two arcade songs in the levels, one for the odd levels and one for the even ones;
- the arcade shop music in the shop and on the `PREPARE` screen;
- the Master Xybot's music and his voice: when the fight starts, when he is hit, and "I'll be back" when he is defeated.

With more than 128 KB of Moonsound sample RAM, shots, hits and some pickups, exits and teleports use arcade sounds. Other effects still come from the MSX.

10. Tips

- Watch the map: the exits are drawn as a square inside a square. Doors need keys, so pick them up on the way.
- Some levels hide a **warp exit**: `WARP BONUS` takes you straight to level 7 with 25 coins, full energy and 75,000 points.
- Keep coins for the shop: `SHOT POWER`, `BETTER SHIELD` and `SLOW ENERGY` help in the long run.
- The special attack clears a crowded corridor, but it costs energy: use it when many robots are in sight.
- In two-player games, being the first one out gives a coin; from level 6 on, do not stand in your partner's line of fire.
- **P** pauses the game at any moment.

11. Credits and rights

- **Xybots** is a game by **Domark (Tengen)**, 1989. The game, its graphics, music and name belong to their rights holders. This is an unofficial port made for study, preservation and homage; it is not affiliated with them.
- **V9990 version** (cartridge version, every picture, panel and text drawn by the V9990, arcade-style graphics, new controls and keyboard for player 2, pause key, Moonsound music and voices, DiHalt logo and cover picture, tools): **DiHalt Studio**, 2026.
- MSX is a trademark of MSX Licensing Corporation. The MSX logo is shown only to say which computers the game runs on (file `MSX-Logo.svg`, Wikimedia Commons).

- V9990 is the name of the Yamaha video chip; the chip photo is by Yaca2671, CC BY-SA 3.0.
- **Xybots** was first an arcade game by **Atari Games**, 1987. The title screen of the MSX version credits `Z80 BY B. COSTAS` and `68000 BY J.TRIPP`.
- With a Moonsound, the music and voices are those of the arcade game, adapted; the song of the logo screen is by DiHalt Studio.
- Source code available on GitHub. **Please improve the game if you feel so!**

Appendix. Cartridge label

A label for the cartridge, ready to print. At 100 % it is about 16 cm wide; scale it in your printing program to fit the label area of your cartridge (the artwork has a 3:2 proportion).

